HECTOR GENARO PACHECO

Software Engineer | Backend • Cloud Infrastructure • DevOps

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PROFESSIONAL SUMMARY

Backend and cloud-focused software engineer with experience across standalone cloud architectures and collaborative team development in shared codebases. Hands-on experience building containerized REST & GraphQL microservices, relational PostgreSQL/Prisma schemas, AWS ECS Fargate deployments, zero-secret runtime configuration (AWS Secrets Manager/KMS), Jira-driven ticket workflows, Git/GitLab version control, and automated testing.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, SQL, Bash/Shell

Cloud & Infra: AWS (ECS Fargate, ECR, RDS, Secrets Mgr, KMS, VPC, IAM), Vercel

Databases: PostgreSQL, Prisma ORM, Neon, Supabase, Relational Modeling, Indexing

Architecture: Microservices, API Contract Design, Data Isolation, Deterministic Math

Backend & APIs: Node.js, Express, Twilio Voice API, REST APIs, GraphQL, Apollo Server, FastAPI, Zod, RBAC

Containers & OS: Docker, Multi-Stage Builds, Container Registries (ECR), Linux, systemd

DevOps & Testing: Git, GitLab, GitHub, Jira, Docker CLI, Vitest, Integration Testing, Build Verification

Security & Auth: JWT Authentication, HTTP-only Cookies, Secrets Injection, Human-in-the-Loop Approval Gates

PROFESSIONAL & TECHNICAL EXPERIENCE

Pacific Horizon Labs LLC — *Founder & Software Engineer* COMPANY SITE ↗ RISING OPS ↗ PHOTO BOOTH ↗ **2026 – Present**

- Architected Pacific Rising Ops, an operations automation service integrating Twilio Voice and Gmail API draft intelligence, enforcing a deterministic finite state machine where protected external actions strictly require explicit human voice approval.
- Engineered cloud backend services and media asset storage pipelines on Supabase with dynamic signed token generation and automated retention cleanup.
- Designed zero-trust approval gates and immutable audit logging for sensitive business operations, ensuring safe AI boundary enforcement.

Instrumentation & Controls Technician — *Process Control & Industrial Systems* **Field Engineering Experience**

- Maintained high-availability industrial control infrastructure, diagnosing electrical loops, PLC I/O wiring, relays, and field instrumentation across 480VAC, 120VAC, and 24VDC systems with systematic root-cause isolation.

Software Engineering Contributor — *Nonprofit serving underserved communities* **Collaborative Contribution**

- Collaborated on an existing nonprofit codebase using Jira-driven ticket workflows, Git/GitLab version control, API integration investigation, local environment configuration, and shared build and test workflows.

FEATURED SOFTWARE ENGINEERING PROJECTS

CalTrack — *Industrial Calibration SaaS & API Platform* (Jun 2026 – Jul 2026) **Node.js • Express • TypeScript • PostgreSQL • Prisma • Docker • AWS ECS Fargate • ECR** LIVE APP ↗ GITHUB ↗

- Architected and deployed containerized Node/Express REST API on AWS ECS Fargate with Amazon RDS PostgreSQL in isolated VPC subnets.
- Built multi-stage Docker pipeline compiling shared TypeScript packages, generating Prisma client, running automated tests, and building linux/amd64 images.
- Configured Amazon ECR image hosting and injected runtime environment credentials securely using AWS Secrets Manager and AWS KMS.
- Enforced role-based access control (RBAC), database foreign-key constraints, and historical snapshot preservation for compliance auditing.

EstateOS — *Agentic Real Estate Investment Intelligence Platform* (Aug 2026 – Sep 2026) **TypeScript • Node.js • Next.js 16 • Claude API • PostgreSQL • Prisma ORM • Vitest** CASE STUDY ↗

- Architected multi-agent orchestration service integrating Claude API research with deterministic underwriting engines and real-time SSE progress streaming.
- Engineered server-side evidence snapshot repositories in PostgreSQL/Prisma, enforcing source-authority tiers and cryptographic deduplication to isolate trusted data.
- Validated deterministic financial invariant calculations, agent workflows, and database repository transactions with automated Vitest suites.

Procurement Tracking System — *Enterprise GraphQL Platform* (May 2026 – Jul 2026) **Node.js • Express 5 • GraphQL • Apollo Server 4 • Prisma ORM 6 • PostgreSQL • JWT** LIVE APP ↗ GITHUB ↗

- Designed and deployed a type-safe GraphQL purchasing API on Render with PostgreSQL on Neon, modeling complex purchase-order and multi-stage receiving state transitions.
- Implemented Apollo Server 4 query/mutation resolvers with JWT authentication, HTTP-only session cookies, and multi-role RBAC resolver permission guards.

Additional Repositories & Deployments: **FieldTrack AI** — Edge Telemetry (Raspberry Pi/Python/React) GITHUB ↗ • **OC Water Features** — Production Web Platform (Next.js/Supabase/Vercel) LIVE SITE ↗ • **Pi Arcade OS** — Embedded Linux System (Python/GPIO) SITE ↗ GITHUB ↗

EDUCATION & PROFESSIONAL TRAINING

Western Governors University (WGU) — B.S. Software Engineering (In Progress, 2026–Present)
Palantir Technologies — Foundry & AIP Learning Credentials (May 2026)
Mirion Technologies — SU-890 iCAM Operations & Maintenance (March 2026)